

Data Science

CS301

The Vaccine Lab

Week 11

Fall 2025

Oliver BONHAM-CARTER

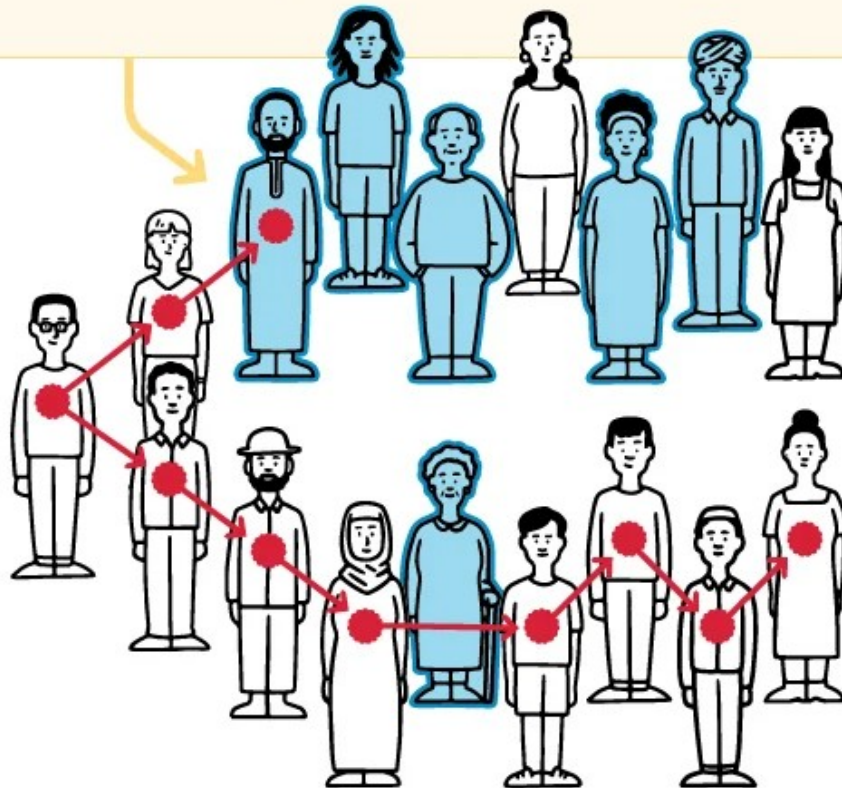
Let's Talk About Lab 5 For A Moment...

- How do you know if something to prevent sickness is working?
- Are the Vaccines working?
 - Are there fewer people with Measles, mumps, Hepatitis B (and other illnesses) as a result of receiving vaccines in 1966?
- History of Vaccines: <https://historyofvaccines.org/>

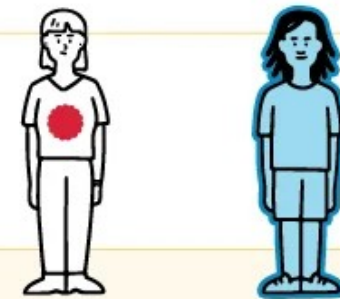


Why Vaccines?

Vaccines do not provide full (100%) protection, so breakthrough infections can happen.



But as more people get vaccinated, it is expected fewer people will come into contact with the virus.



INFECTED

VACCINATED



What Does Research Say?

Comparison of 20th Century Annual Morbidity & Current Morbidity

Disease	20 th Century Annual Morbidity*	2010 Reported Cases [†]	% Decrease
Smallpox	29,005	0	100%
Diphtheria	21,053	0	100%
Pertussis	200,752	21,291	89%
Tetanus	580	8	99%
Polio (paralytic)	16,316	0	100%
Measles	530,217	61	>99%
Mumps	162,344	2,528	98%
Rubella	47,745	6	>99%
CRS	152	0	100%
<i>Haemophilus influenzae</i> (<5 years of age)	20,000 (est.)	270 (16 serotype b and 254 unknown serotype)	99%

Sources:

* JAMA. 2007;298(18):2155-2163

† CDC. MMWR January 7, 2011;59(52);1704-1716. (Provisional MMWR week 52 data)

- Vox Article: <https://www.vox.com/health-care/2014/10/13/6967317/vaccines-work-this-chart-proves-it>



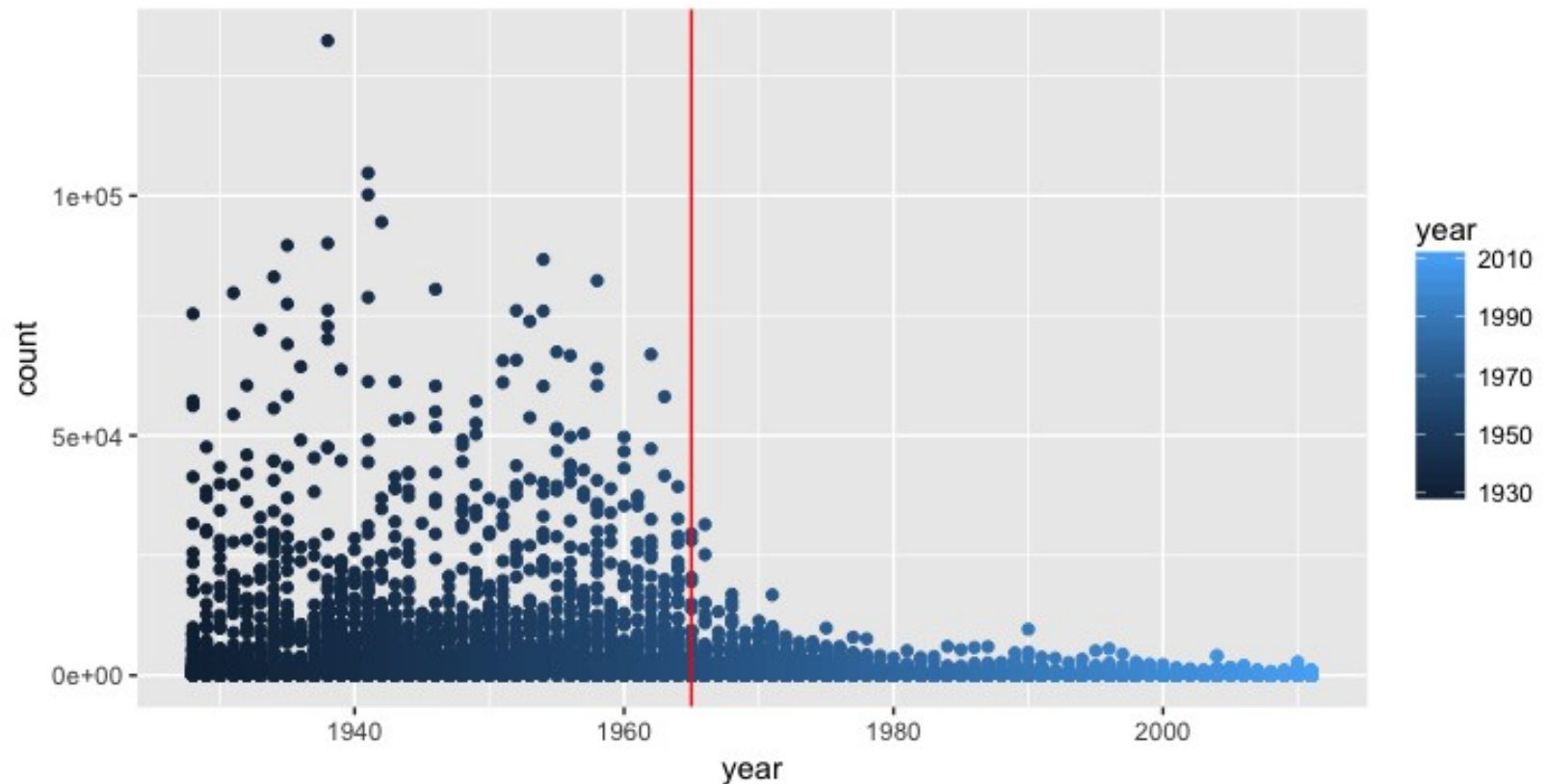
What Does **Our Data** Say About (All) Vaccines of Data?

```
library(tidyverse)
```

```
library(dslabs)
```

```
library(dplyr)
```

```
ggplot(data = us_contagious_diseases) + geom_point(mapping = aes(x = year,  
y = count, color = year)) + geom_vline(xintercept = 1965, color = "red")
```



Cases
of
Illness



Lab Results

- Use the us contagious disease and dplyr tools to create an object that **stores only the Measles data**, **includes a per 100,000 people rate**, and removes Alaska and Hawaii. **Note that there is a weeks reporting column. Take that into account when computing the rate.**

- # Add the rate column to the data:

```
dat_measles_rate <- filter(us_contagious_diseases, disease ==  
"Measles") %>% mutate(rate = (((count*52)/weeks_reporting)) /  
((population)/100000))
```

- # Note: the *rate* is one of several possible calculations...



Trim Out Two States

```
# Remove the two states (Alaska and Hawaii)
```

```
dat_measles_rate_lessTwoStates <- filter(dat_measles_rate,  
state != "Alaska", state != "Hawaii")
```

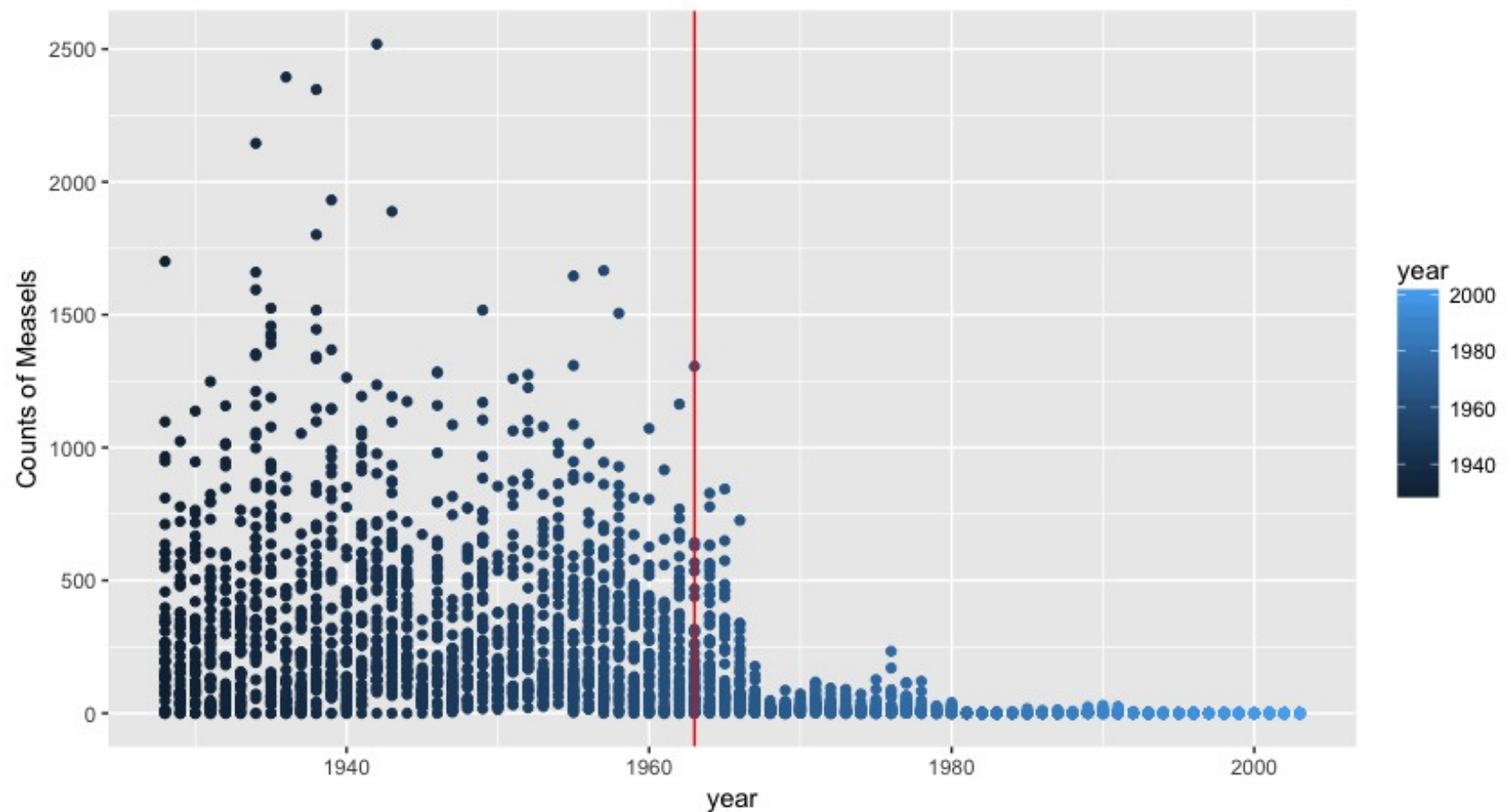
```
View(dat_measles_rate_lessTwoStates)
```

```
# Plot the results across 48 states
```

```
ggplot(data = dat_measles_rate_lessTwoStates, mapping =  
aes(x = year, y = rate, color = year)) + geom_point() +  
geom_vline(xintercept = 1963, color = "red") + labs(y = "Counts  
of Measels")
```


Plot Across 48 States

```
ggplot(data = dat_measles_rate_lessTwoStates, mapping = aes(x =  
year, y = rate, color = year)) + geom_point() + geom_vline(xintercept  
= 1963, color = "red") + labs(y = "Counts of Measels")
```





Focus On California

```
# Create table to focus on California
```

```
dat_california <- filter(dat_measles_rate_lessTwoStates, state  
== "California")
```

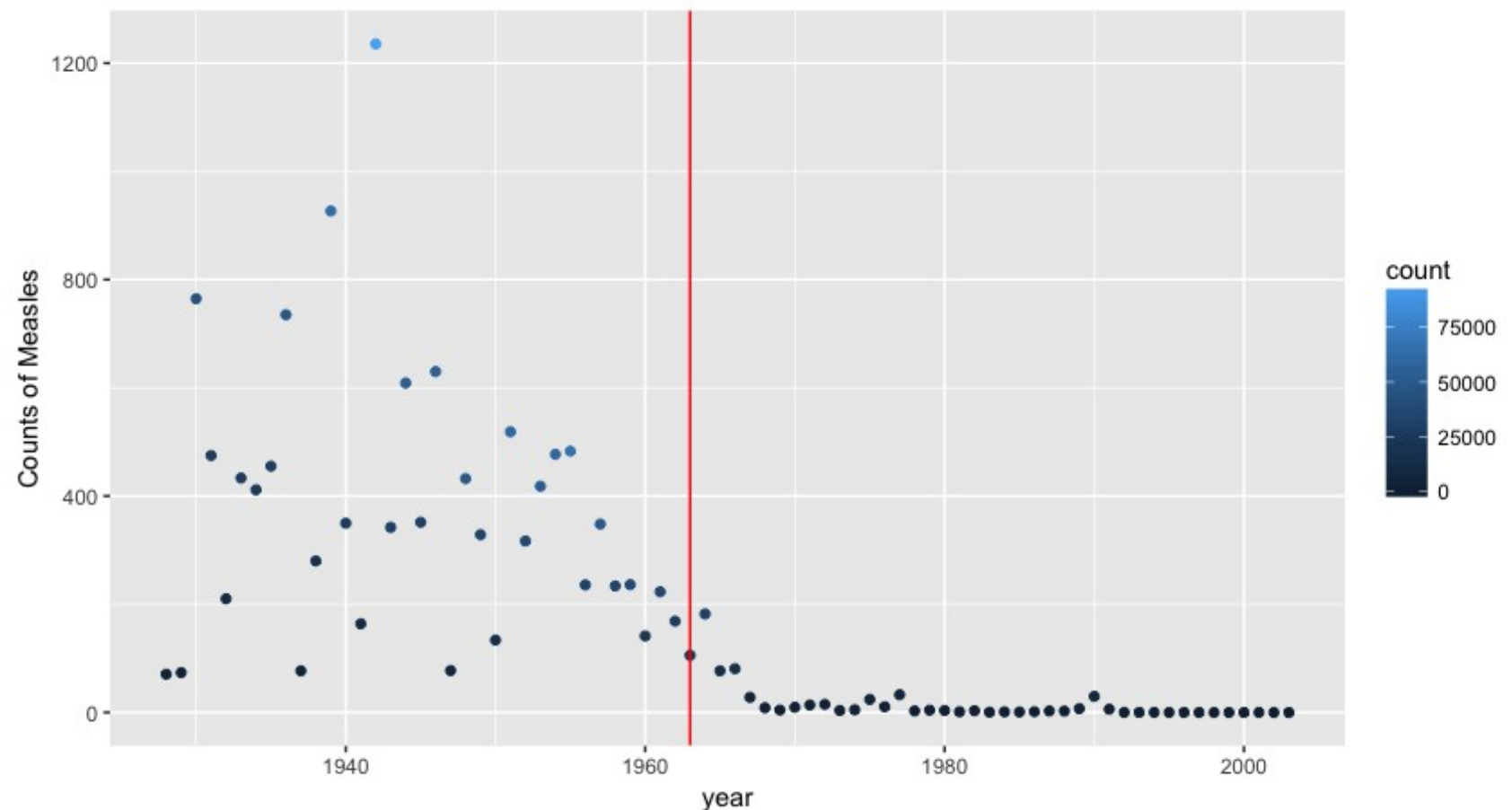
```
View(dat_california)
```

```
ggplot(data = dat_california, mapping = aes(x = year, y = rate,  
color = count)) + geom_point() + geom_vline(xintercept = 1963,  
color = "red") + labs(y = "Counts of Measles")
```



Data From California, Only

```
ggplot(data = dat_californiaFocus, mapping = aes(x = year, y = rate, color = count)) +  
  geom_point() + geom_vline(xintercept = 1963, color = "red") +  
  labs(y = "Counts of Measles")
```





Discussion Points

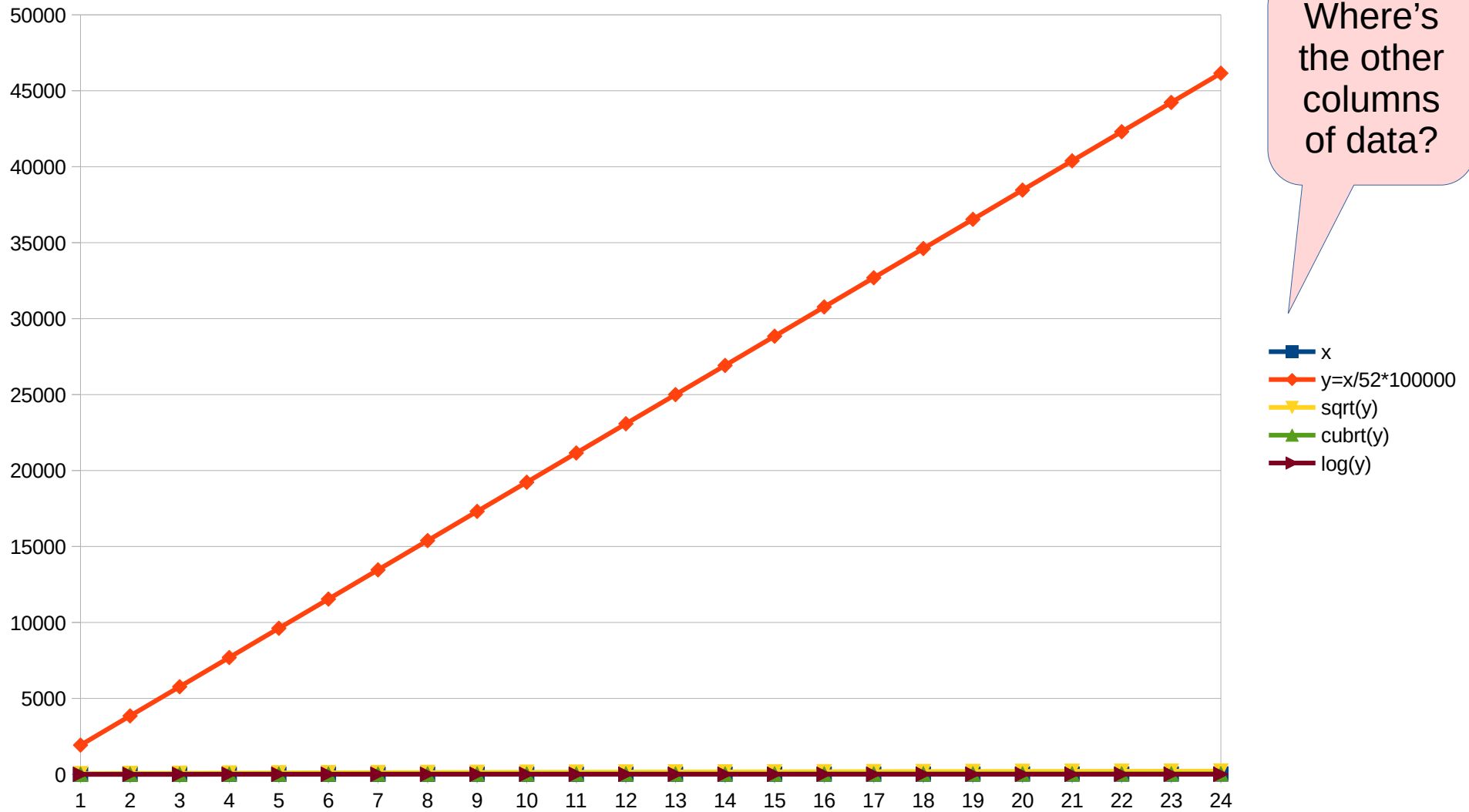
- *Vaccines administered in 1964 cause a decline in measles cases.*
- *California was used as a sample study: less data but still a good representation of the rest of the dataset*
- *The decades following the vaccines of 1964 show that new threats appear. Vaccines must be updated.*



Transformations Helped Visualize

- *Allows us to visually compare data by changing shape*
- The square root, x to $x^{(1/2)} = \text{sqrt}(x)$, is a transformation with a moderate effect on distribution shape.
- Weaker than the logarithm and the cube root transformations
- Used for reducing right skewness
- Has the advantage that it can be applied to zero values

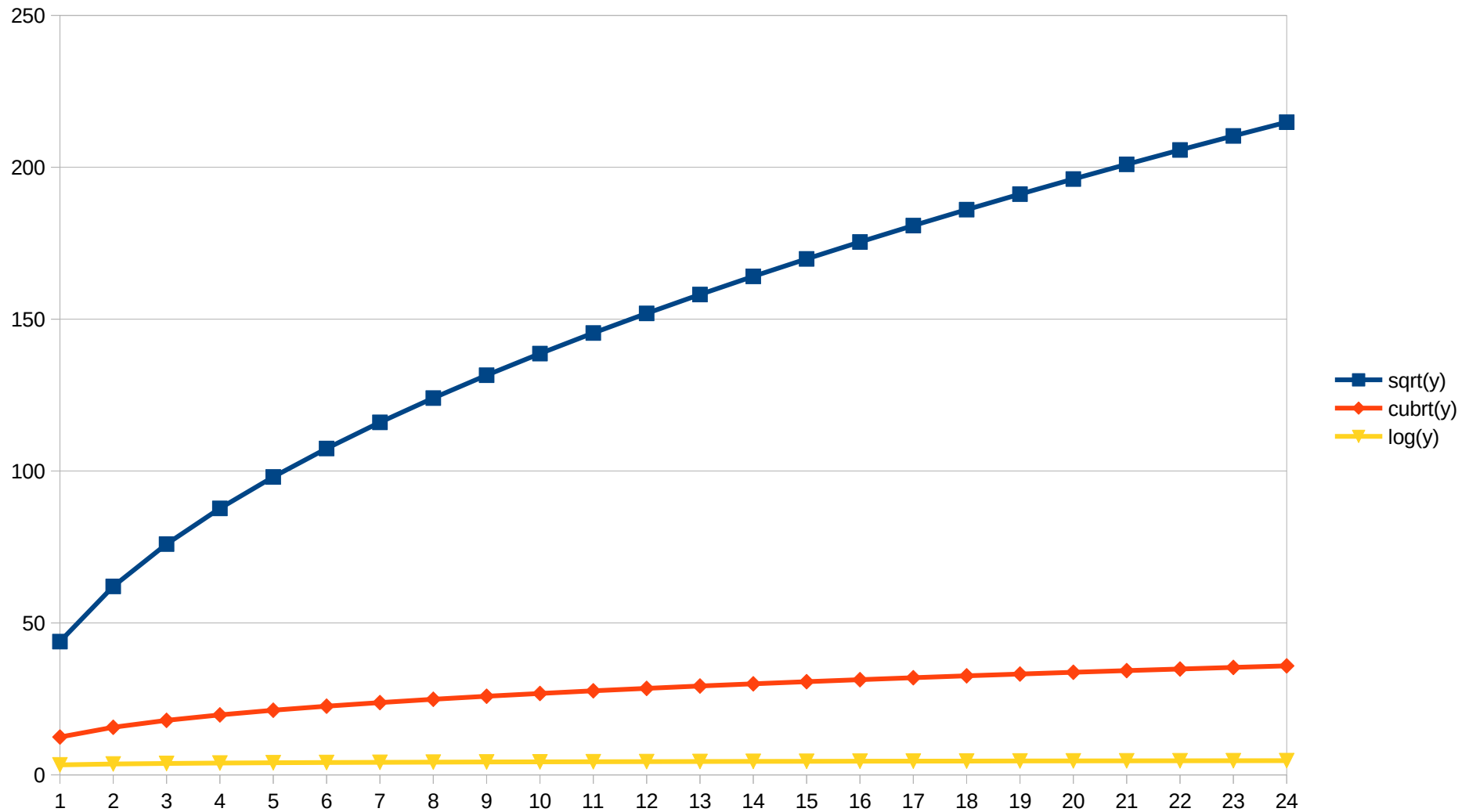
Effects of Transformations on Variables



x	y=x/52*100000	sqrt(y)	cubrt(y)	log(y)
1	1923.076923	43.85290097	12.43556587	3.283996656
2	3846.153846	62.01736729	15.6678312	3.585026652
3	5769.230769	75.95545253	17.93518953	3.761117911
4	7692.307692	87.70580193	19.74023034	3.886056648

Effects of Transformations on Vars

Zoom-in

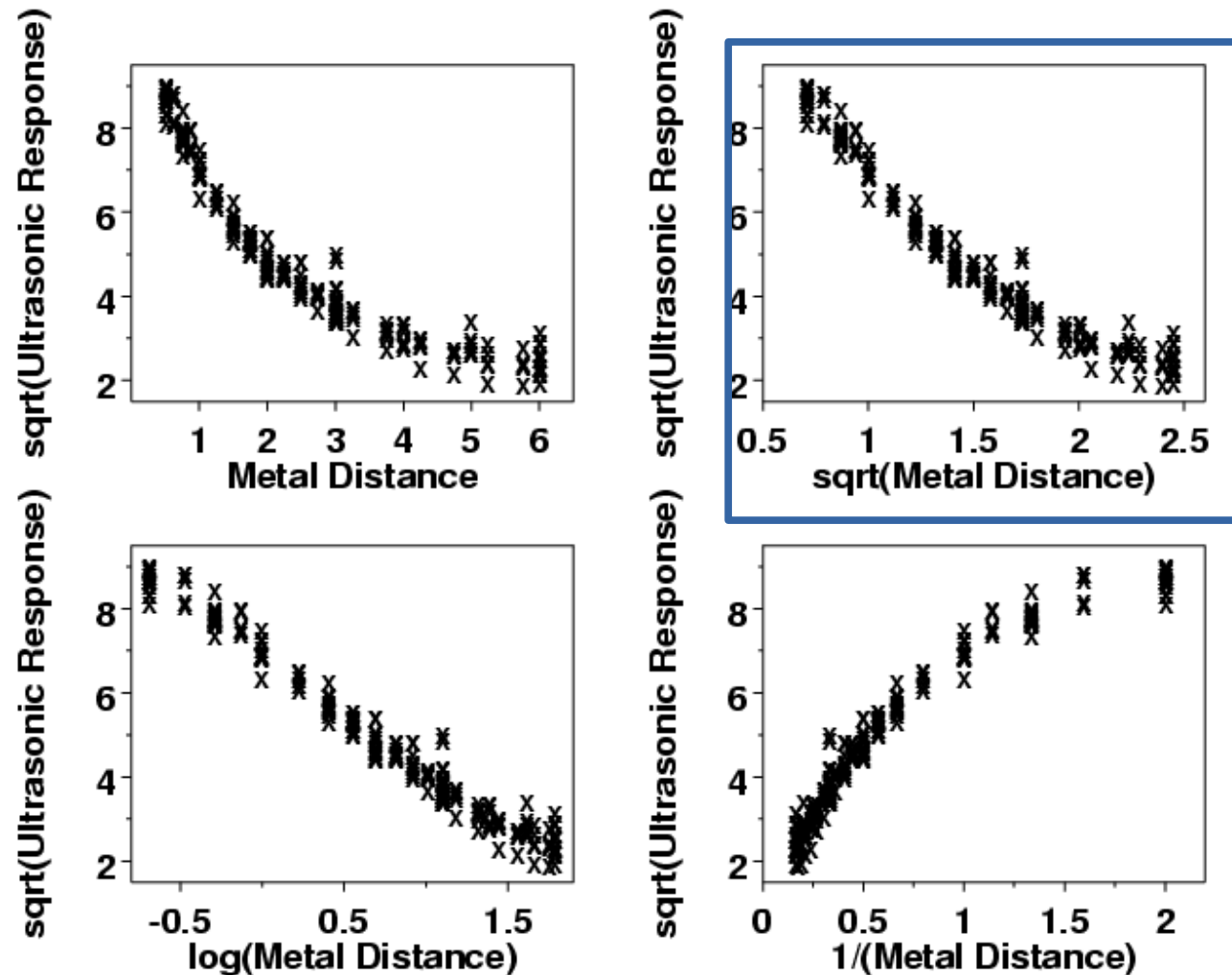




Common Transformations

- Reduce the Y into a smaller space to see trends.
- Places all points on a similar playing ground
- $P \leftarrow (x, y)$
- $\text{Trans}(p) \leftarrow (x, \sqrt{y})$

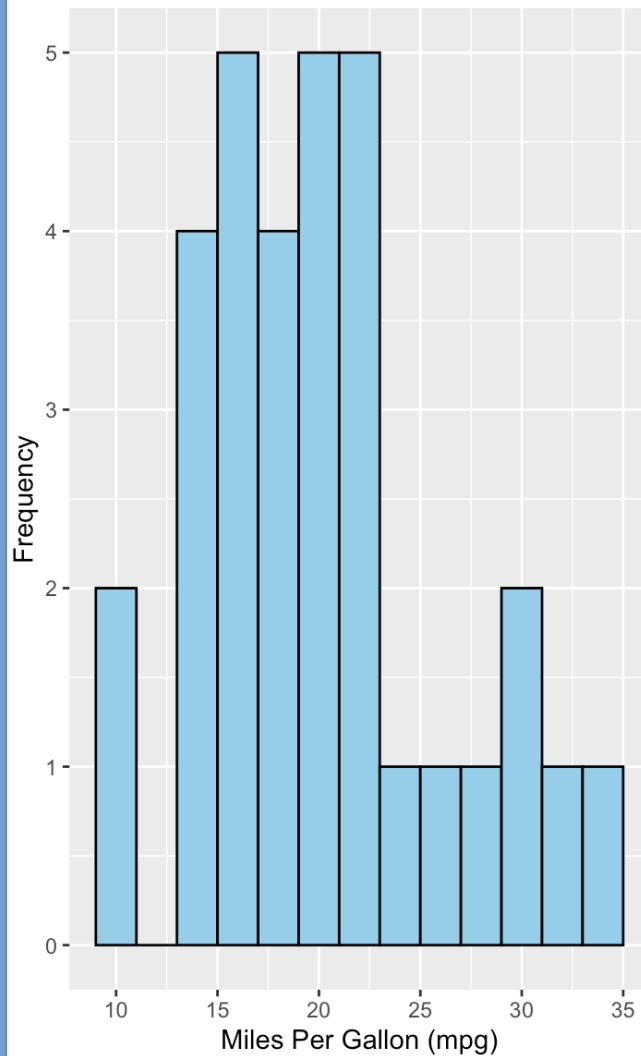
TRANSFORMATIONS OF PREDICTOR VARIABLE



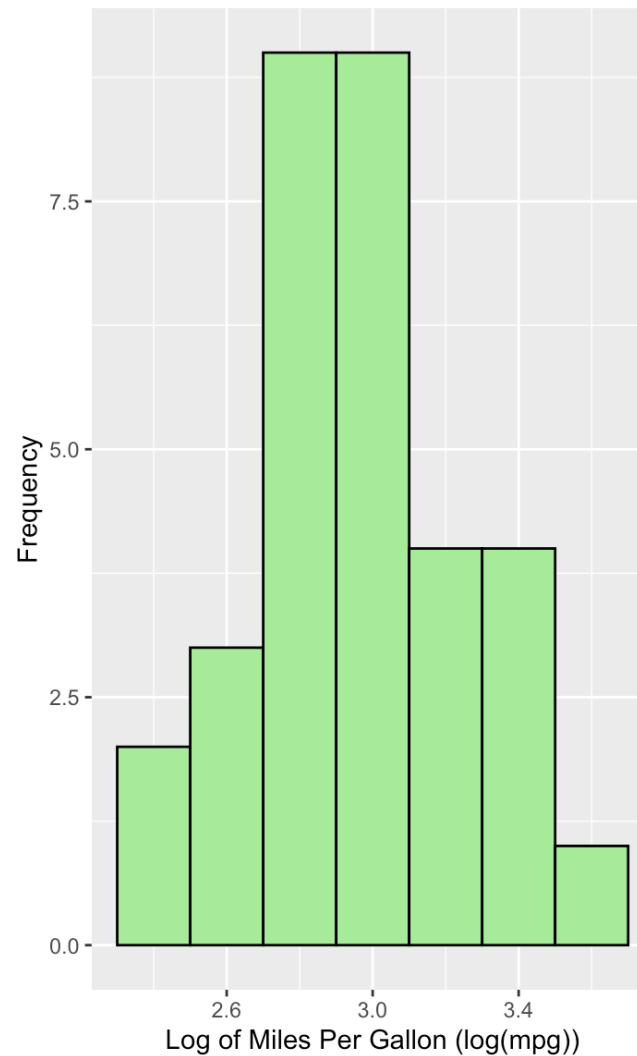


More On Transformations

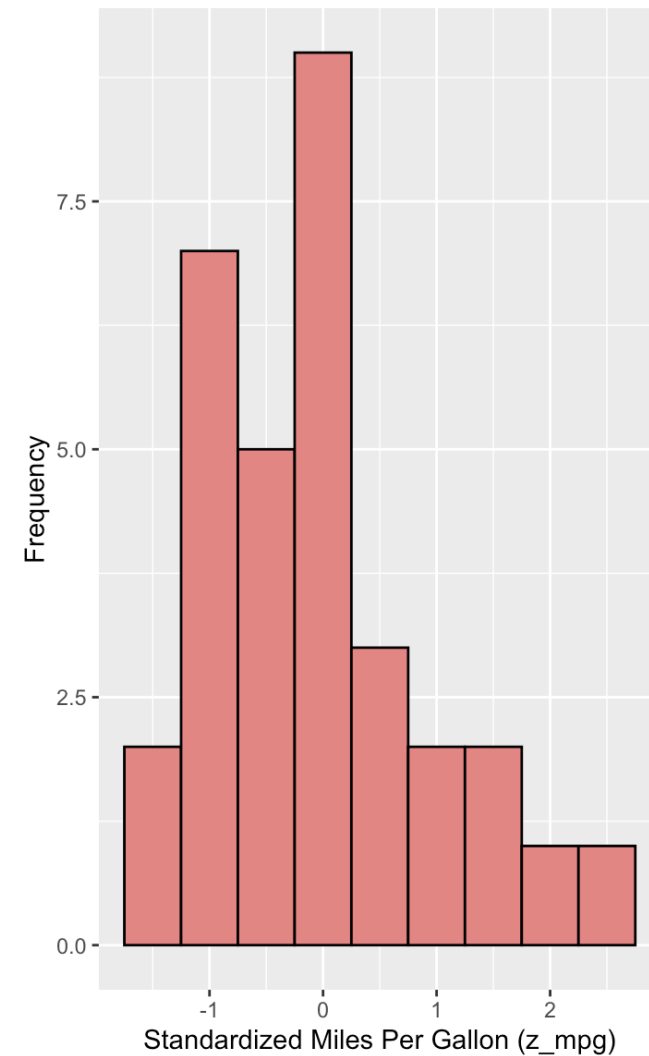
Original mpg Distribution



Log-Transformed mpg Distribution



Standardized mpg Distribution





The 1950's, 1960's and 1970's Without Transformation

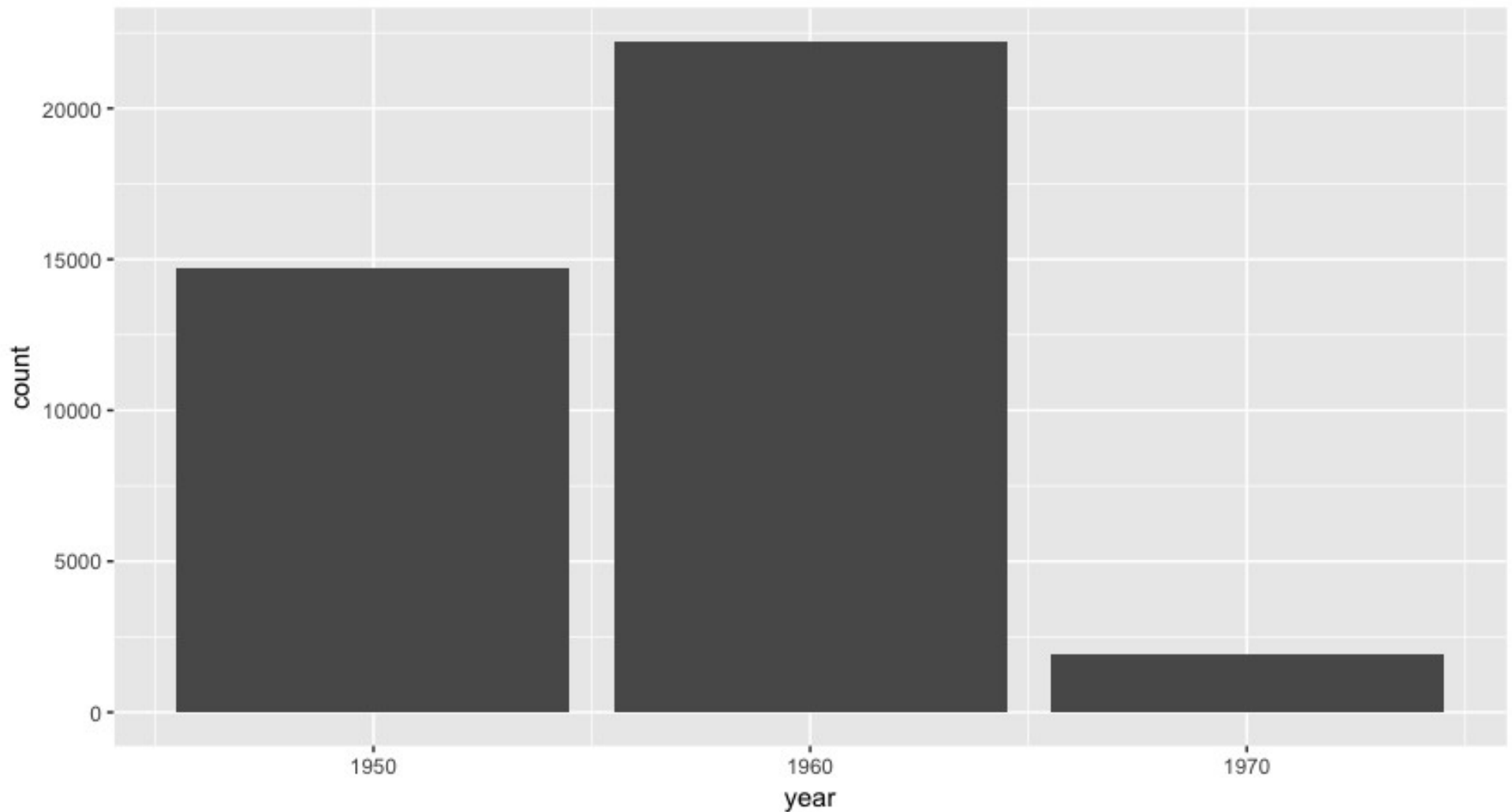
plot three bars to see what happened in the 1950's, 1960's and 1970's.

```
ggplot(data = dat_califocus %>% filter(year ==  
1950 | year == 1960 | year == 1970)) +  
geom_bar(mapping = aes(x = year, y = count),  
stat = "identity")
```

Back to our conversation about vaccines



The 1950's, 1960's and 1970's Without Transformation





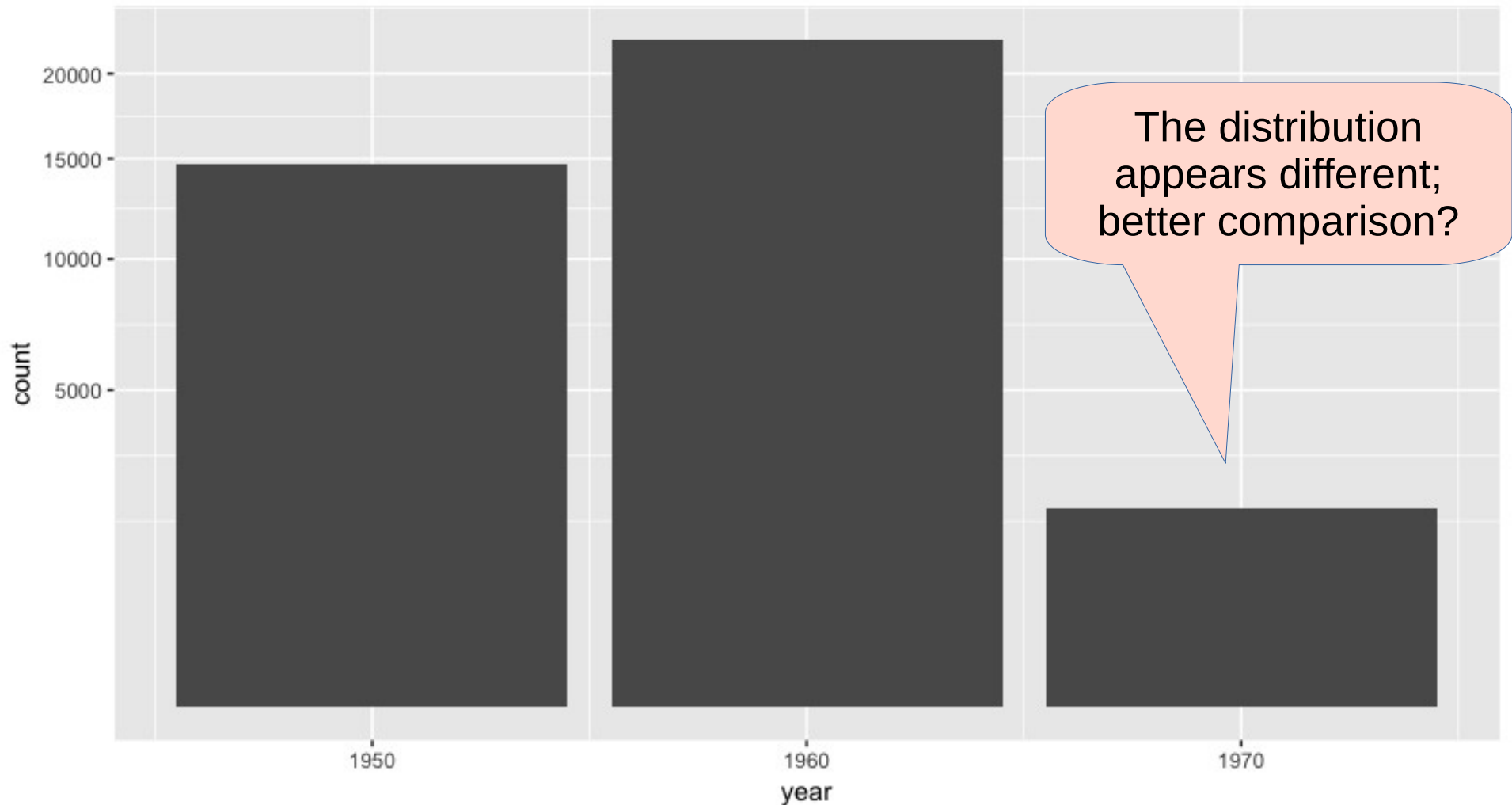
The 1950's, 1960's and 1970's With Sqrt() Transformation

#plot three bars to see what happened in the 1950's, 1960's and 1970's.

```
ggplot(data = dat_caliFocus %>% filter(year ==  
1950 | year == 1960 | year == 1970)) +  
geom_bar(mapping = aes(x = year, y =  
sqrt(count)), stat = "identity")
```



The 1950's, 1960's and 1970's With Sqrt() Transformation





The 1950's, 1960's and 1970's Without Transformation

```
#create some "block", containers to hold the data for each year.
```

```
dat_measles_rate_lessTwoStates$yearBlock[dat_measles_rate_lessTwoStates$year  
== 1950] <- "1950's"
```

```
dat_measles_rate_lessTwoStates$yearBlock[dat_measles_rate_lessTwoStates$year  
== 1960] <- "1960's"
```

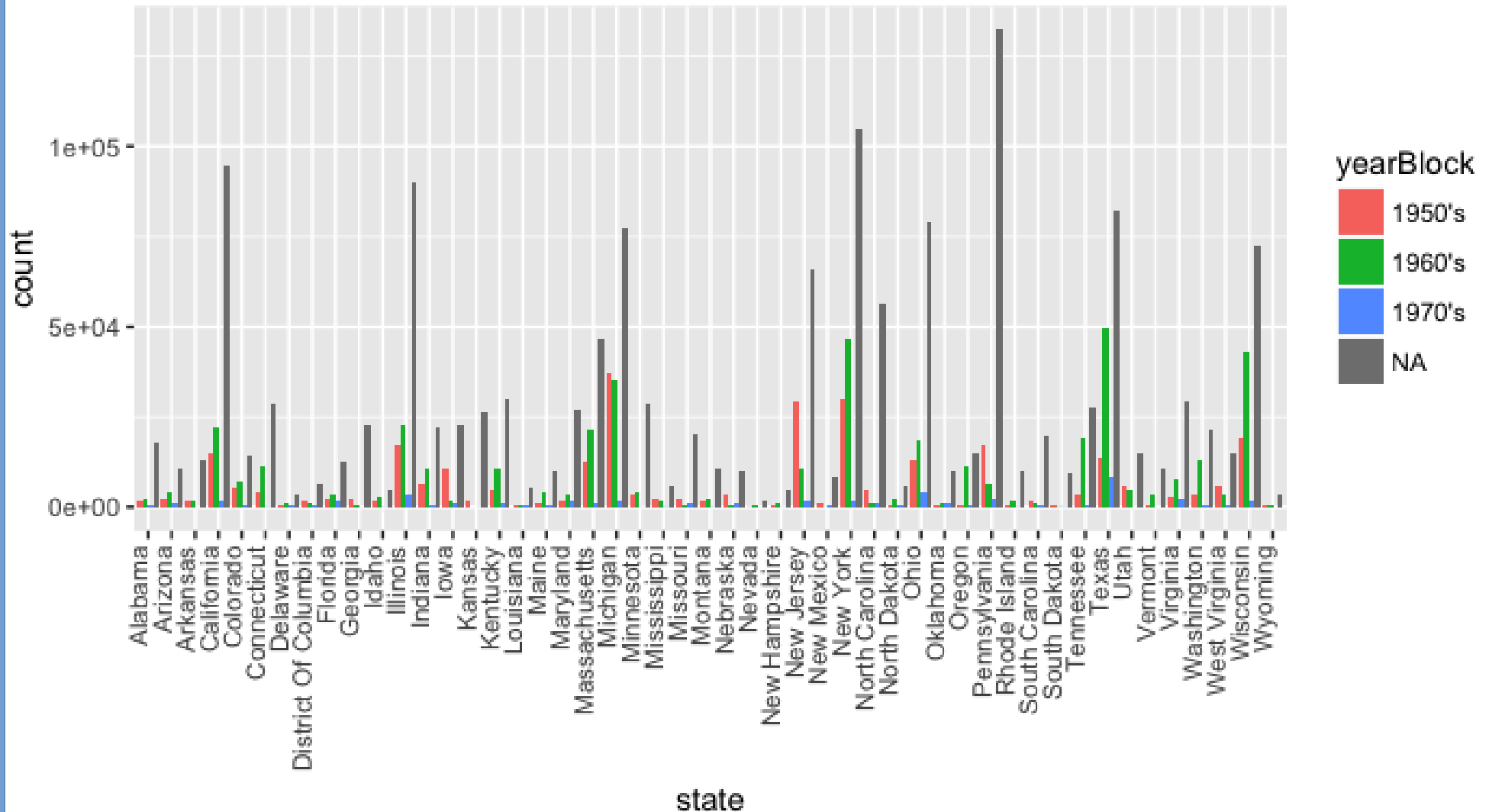
```
dat_measles_rate_lessTwoStates$yearBlock[dat_measles_rate_lessTwoStates$year  
== 1970] <- "1970's"
```

```
#Without transformation, Multi-bar per state,
```

```
ggplot(data = dat_measles_rate_lessTwoStates) + geom_bar(mapping  
= aes(x = state, y = count, fill = yearBlock), position = "dodge", stat =  
"identity") + theme(axis.text.x = element_text(angle = 90, hjust = 1,  
vjust=-0.01))
```



The 1950's, 1960's and 1970's Without Transformation





Comparing States; Two plots

No transformation

```
dat_measles_rate %>%
```

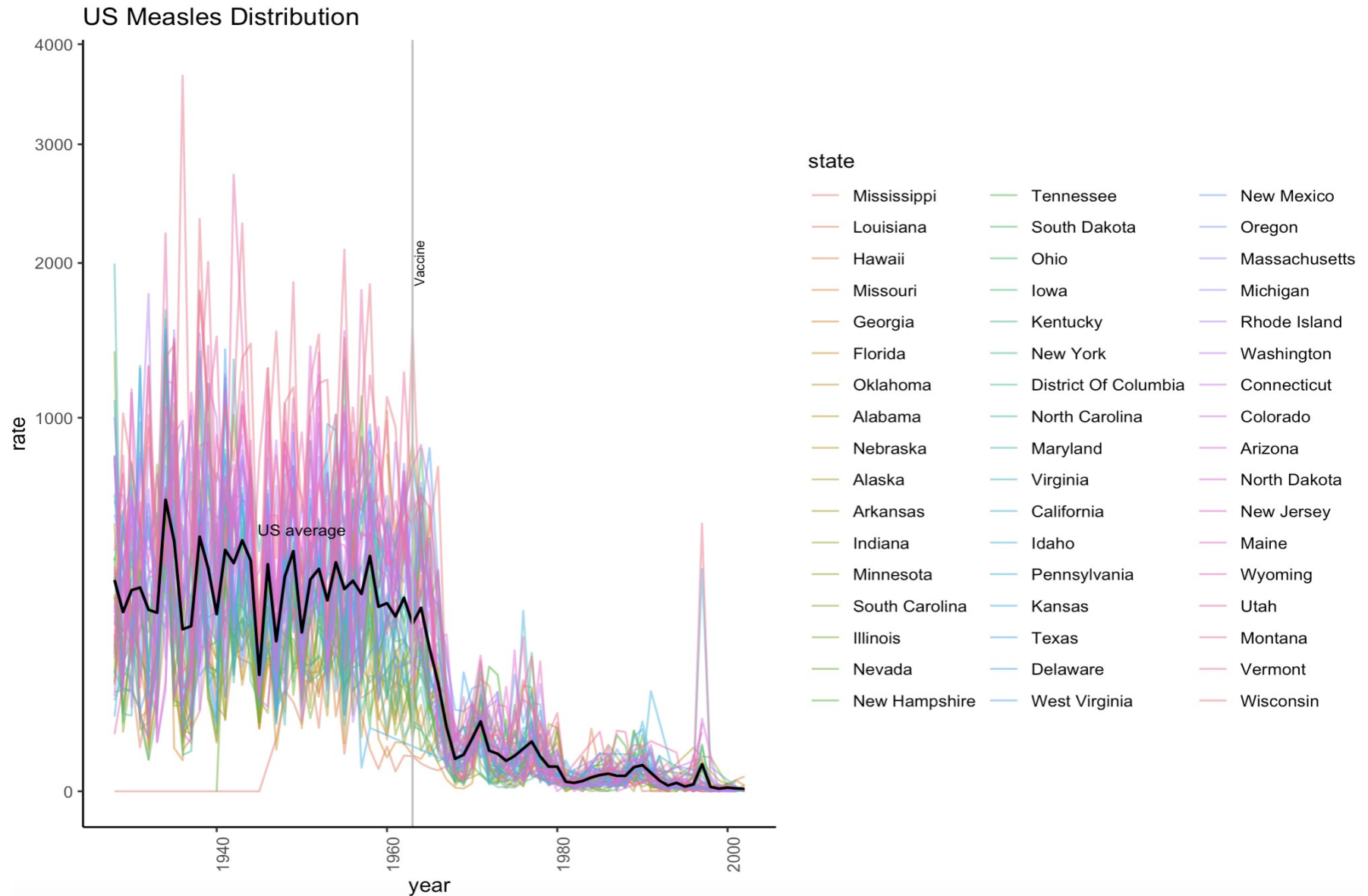
```
  filter(!is.na(rate)) %>% mutate(state = reorder(state, rate, FUN = mean)) %>%  
  ggplot(aes(year, rate, color = state)) + geom_line(alpha = 0.5) + theme_classic() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) + scale_y_sqrt() +  
  stat_summary(fun.y=mean, geom="line", lwd=0.7, col="black") + annotate("text", x =  
  1950, y = 490, label = "US average", size = 3) + ggtitle("US Measles Distribution") +  
  annotate("text", x = 1963, y = 2000, angle = 90, label = paste("paste(Vaccine)", collapse  
  = "_"), vjust = 1.2, parse = TRUE, size = 2.5) + geom_vline(xintercept = 1963, col =  
  "grey")
```

```
dat_measles_rate %>% filter(!is.na(rate)) %>%
```

```
  ggplot(aes(year, state)) + geom_tile(aes(fill = rate), color = "white") +  
  scale_fill_gradient(low = "white", high = "blue", trans = "sqrt") +  
  scale_x_continuous(expand = c(0,0)) + ggtitle("Measles disease rate per year in the  
  United States") + theme(plot.title = element_text(hjust = 0.5)) + geom_vline(xintercept =  
  1963, col = "black")
```

Comparing States

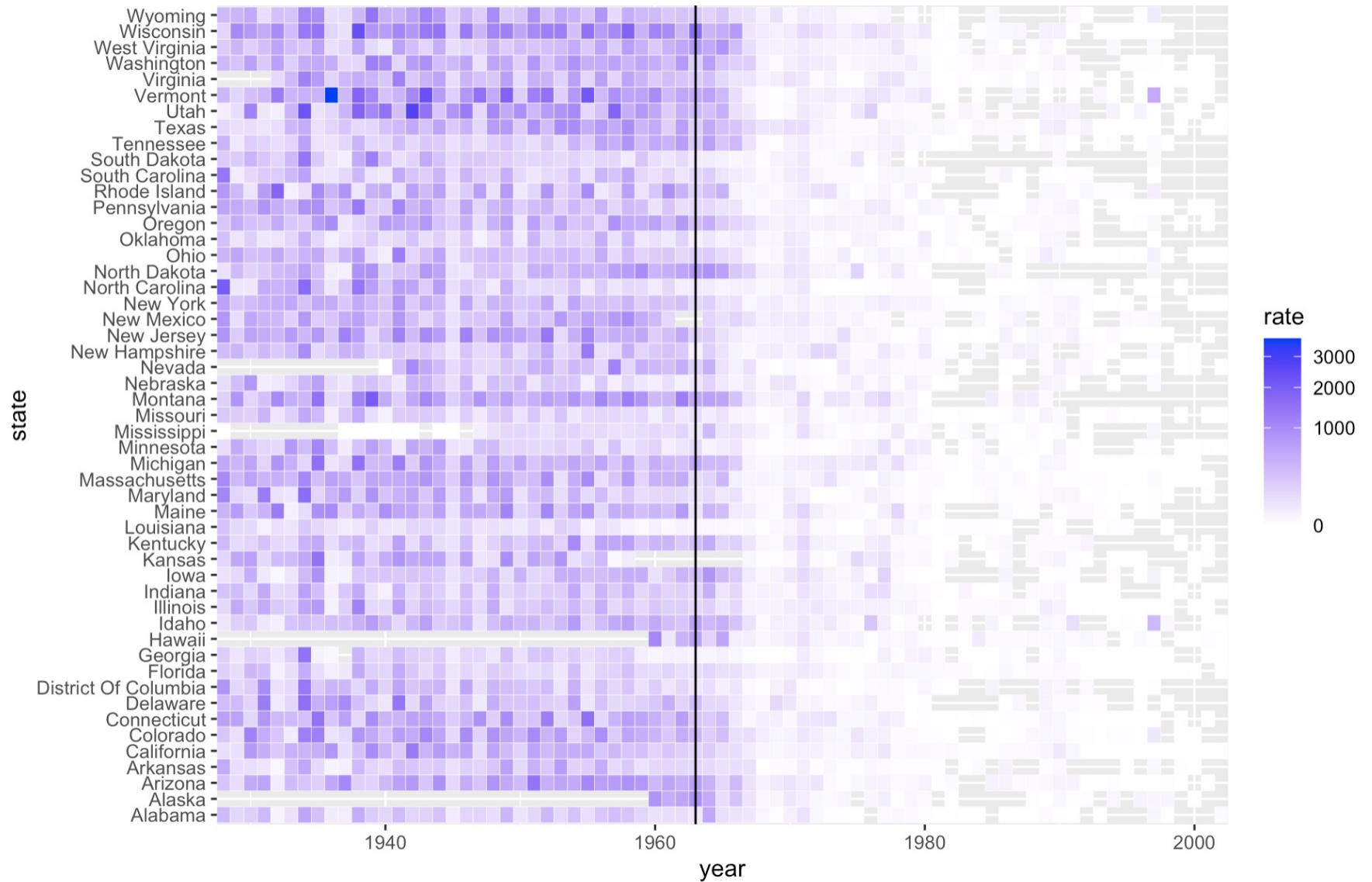
No Transformation



Comparing States

No Transformation

Measles disease rate per year in the United States



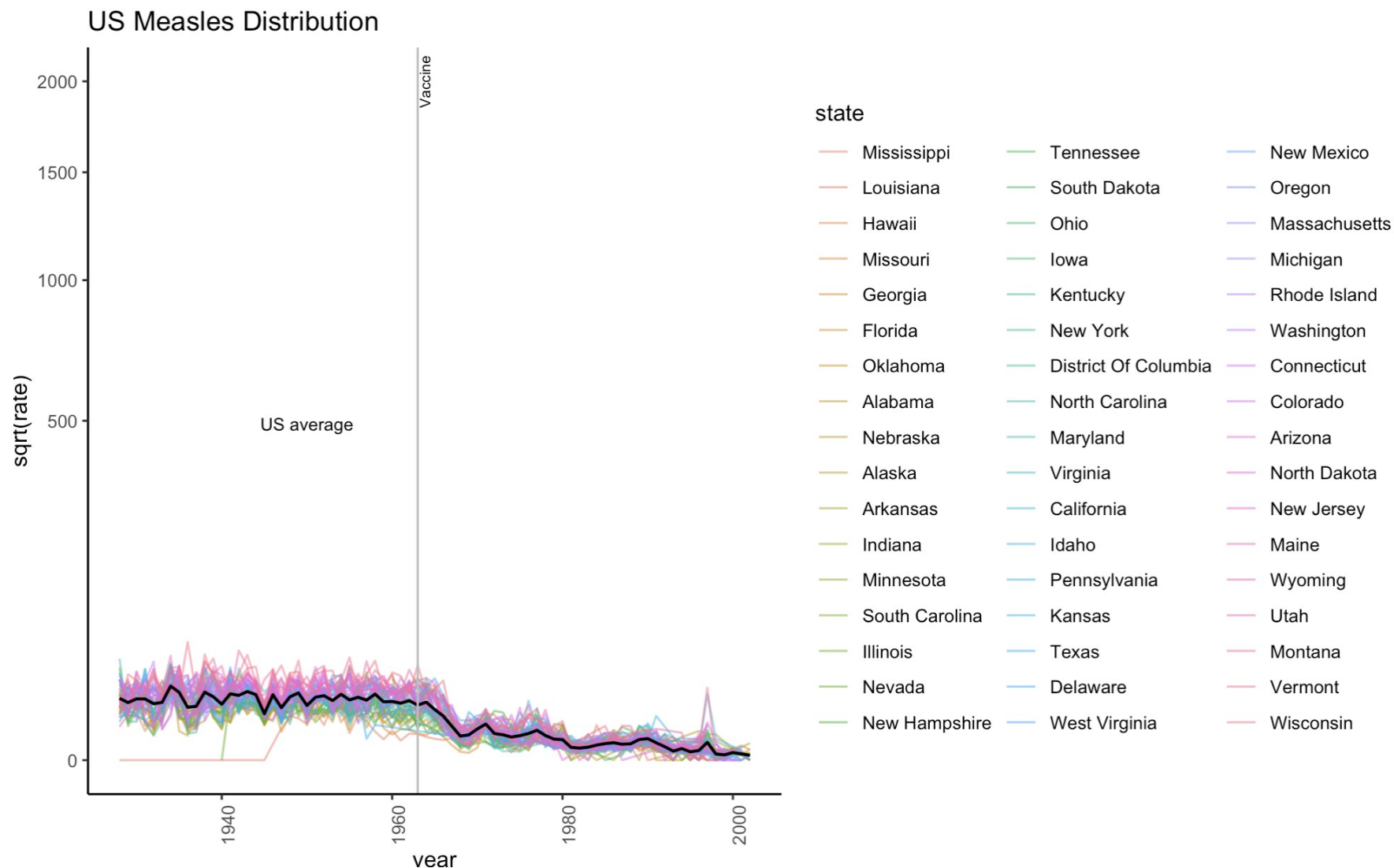


Comparing States With Transformation

```
dat_measles_rate %>%  
  filter(!is.na(rate)) %>% mutate(state = reorder(state, rate, FUN =  
mean)) %>%  
  ggplot(aes(year, sqrt(rate), color = state)) + geom_line(alpha = 0.5) +  
  theme_classic() + theme(axis.text.x = element_text(angle = 90, hjust  
= 1)) +  
  scale_y_sqrt() + stat_summary(fun.y=mean,  
geom="line", lwd=0.7, col="black") +  
  annotate("text", x = 1950, y = 490, label = "US average", size = 3) +  
  ggtitle("US Measles Distribution") +  
  annotate("text", x = 1963, y= 2000, angle = 90, label =  
paste("paste(Vaccine)", collapse = "_"), vjust = 1.2, parse = TRUE, size  
= 2.5) + geom_vline(xintercept = 1963, col = "grey")
```

Does this help compare?

Comparing States With Transformation





Comparing States With Transformation

```
# with transform
```

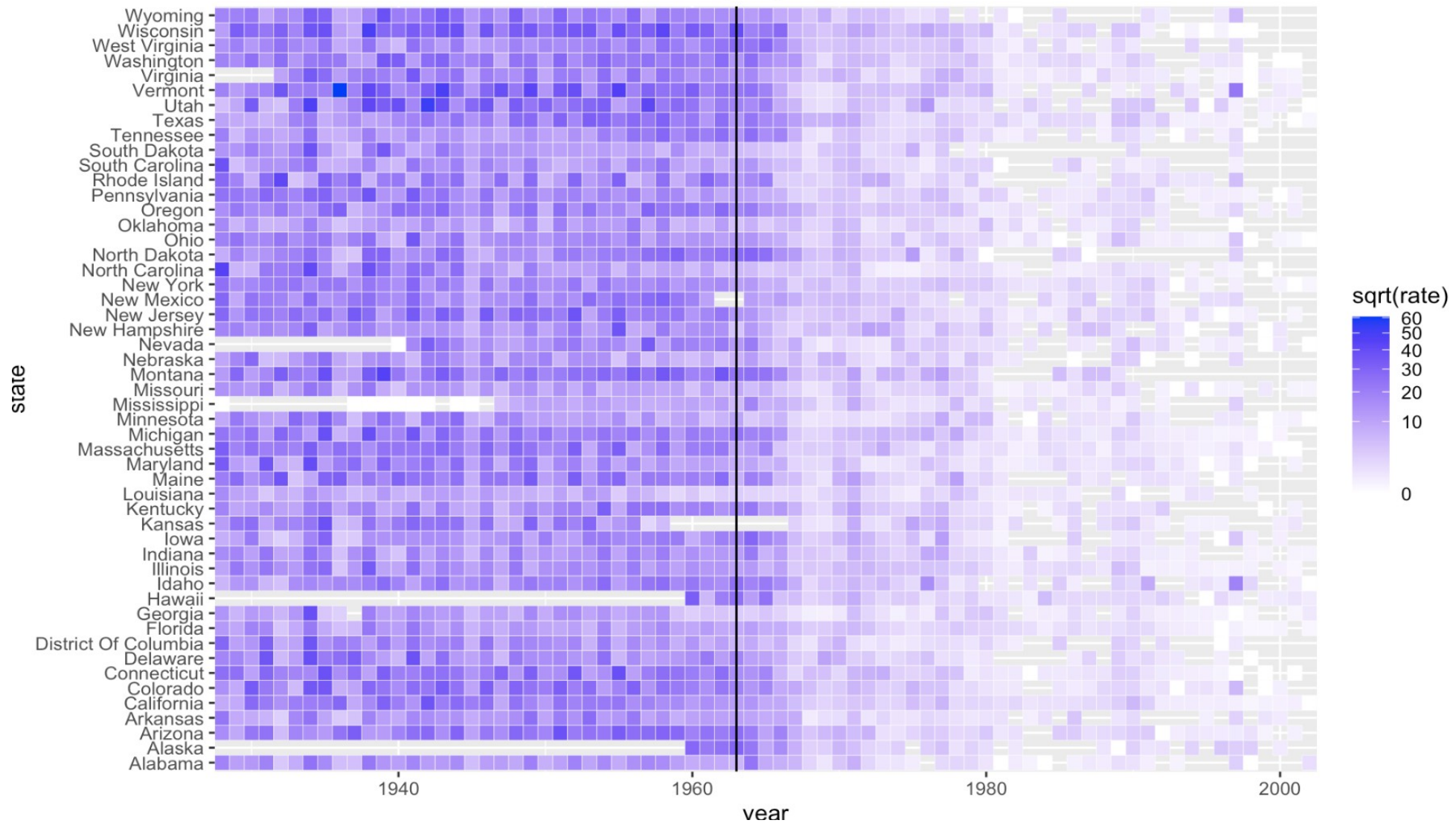
```
dat_measles_rate %>% filter(!is.na(rate)) %>%
```

```
  ggplot(aes(year, state)) + geom_tile(aes(fill = sqrt(rate)),  
    color = "white") + scale_fill_gradient(low = "white", high =  
    "blue", trans = "sqrt") + scale_x_continuous(expand = c(0,0))  
+ ggtitle("Measles disease rate per year in the United  
States") + theme(plot.title = element_text(hjust = 0.5)) +  
geom_vline(xintercept = 1963, col = "black")
```

Does this help compare?

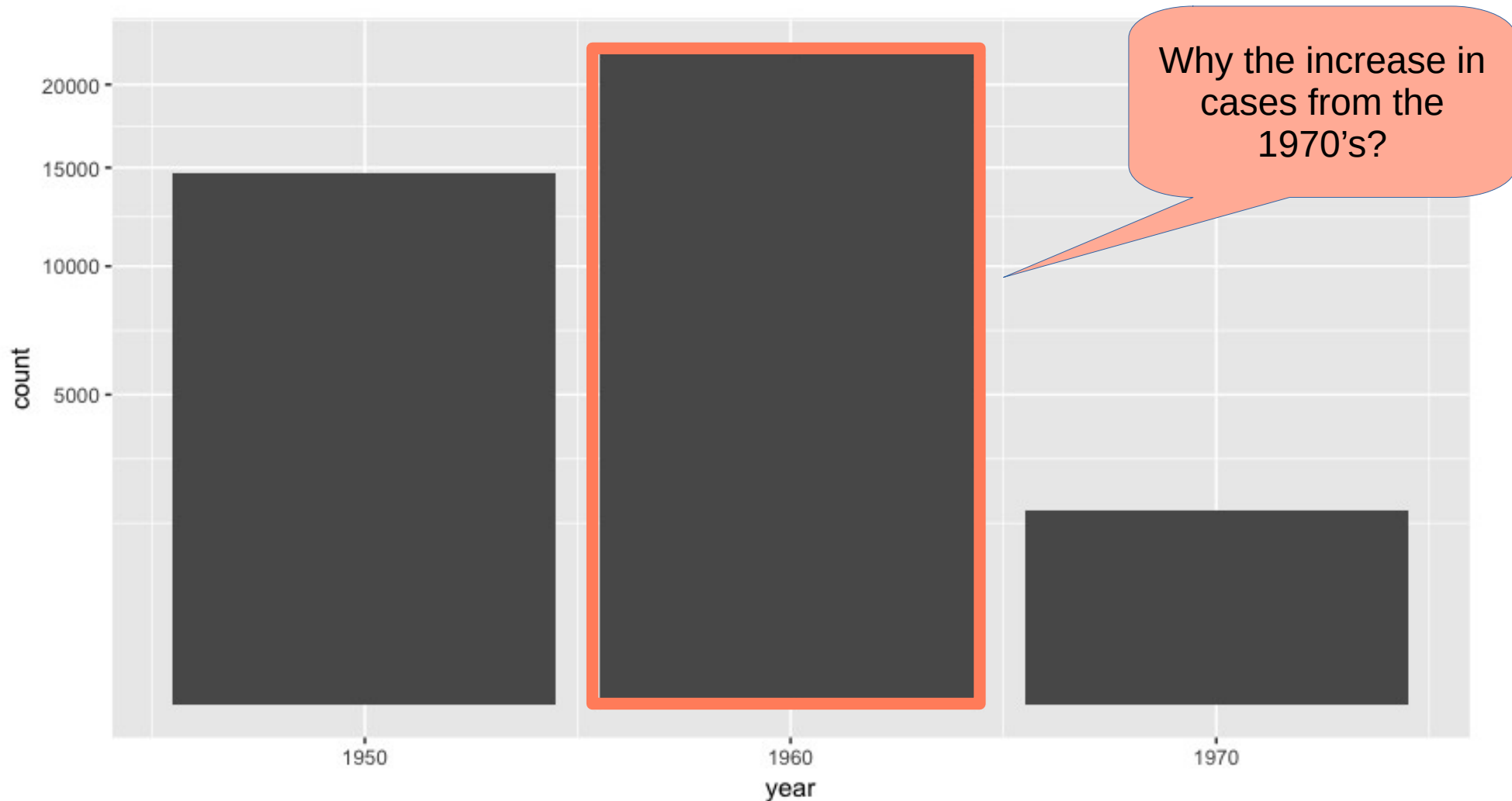
Comparing States With Transformation

Measles disease rate per year in the United States





Going back... Did the 1960's show
an increase in cases?



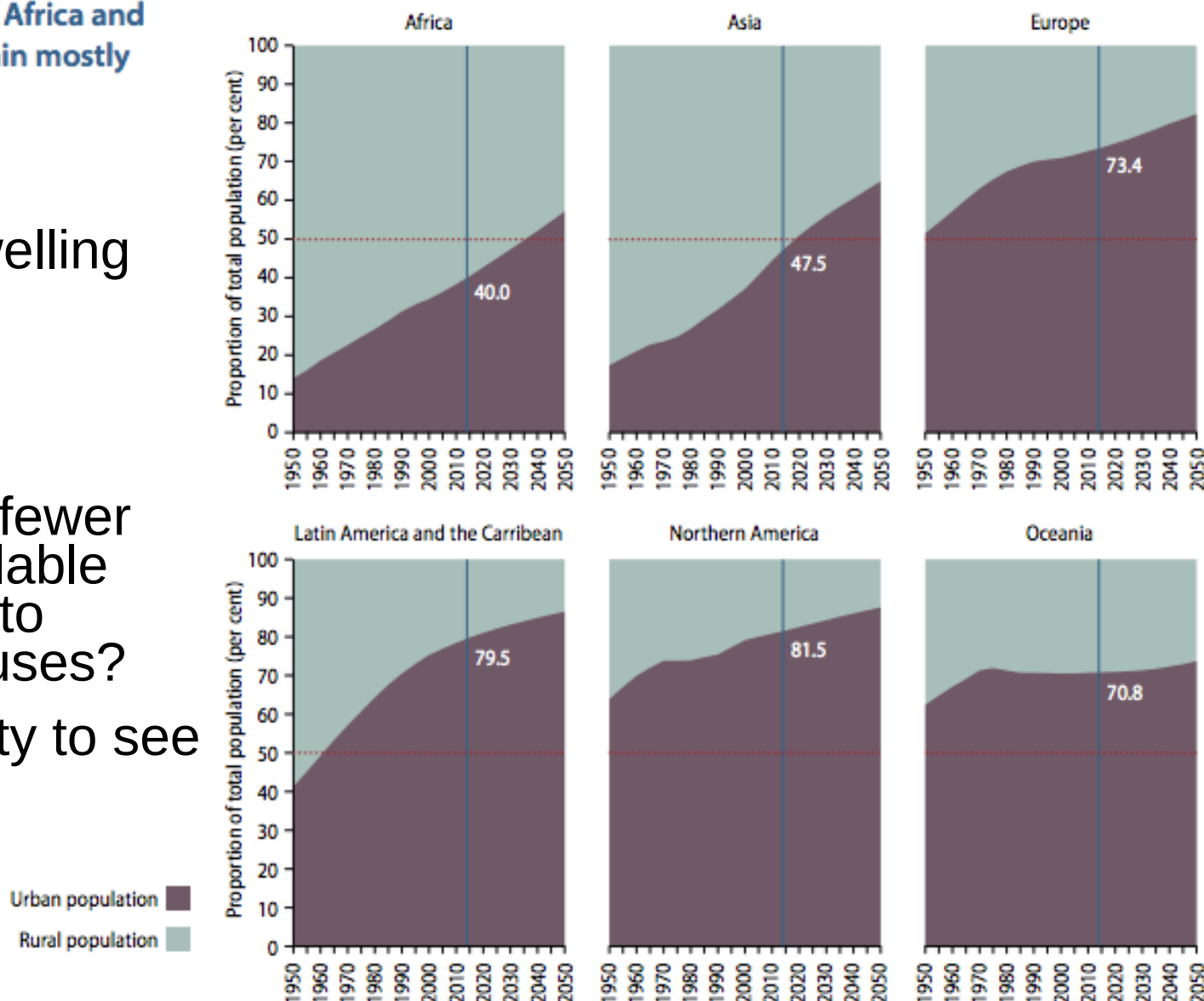
Urban Versus Rural

Urbanization has occurred in all major areas, yet Africa and Asia remain mostly rural

- **Urban:** City dwelling
- **Rural:** Country dwelling
- Vaccinations:
 - Were there fewer people available from whom to contract viruses?
- Less opportunity to see others?

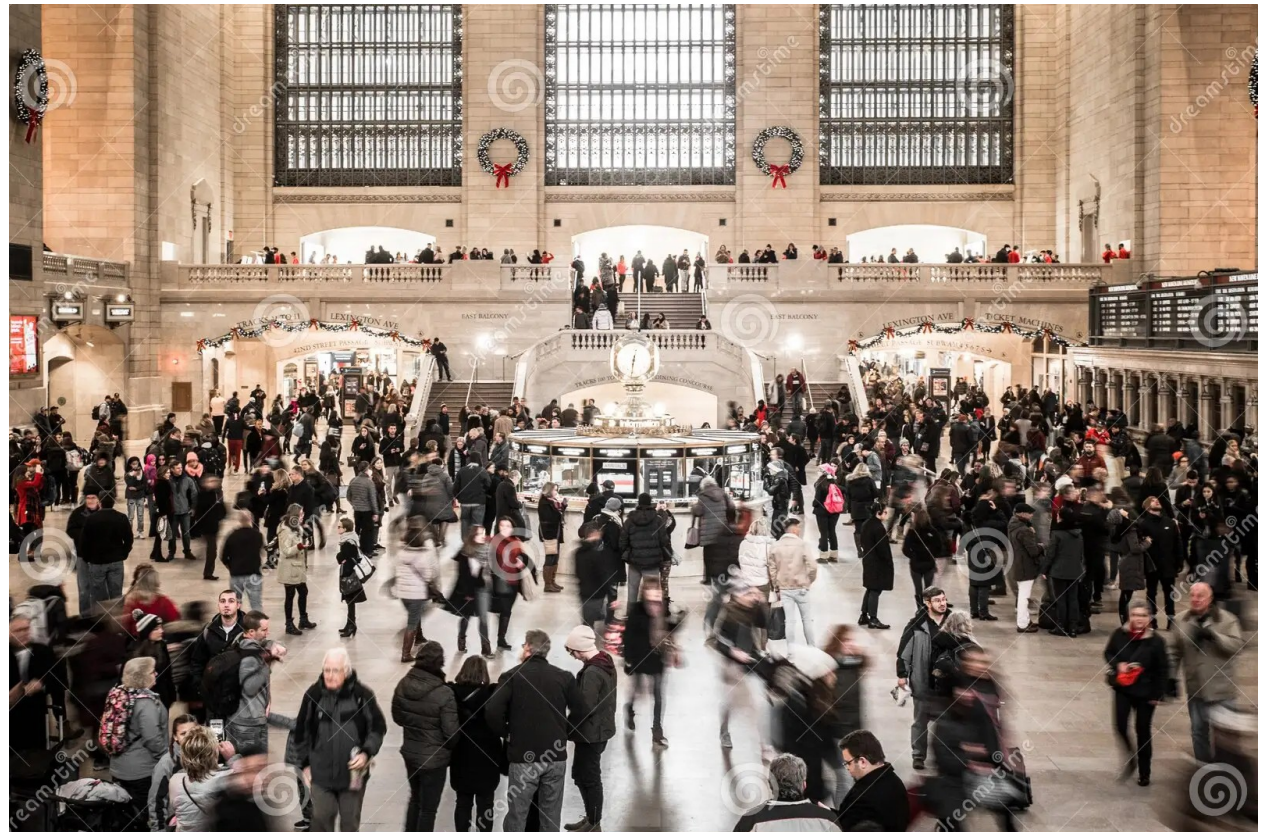
Figure 3.

Urban and rural population as proportion of total population, by major areas, 1950–2050



More People, More Contact

Could a crowding of cities in the 1960's create a better place for measles to spread?



As good as this explanation seems, we still need to verify it by facts.

Concluding Ideas

Why is this *also* important?

Open and Transparent: By making the data about vaccines available to the general public, anyone can study the data to resolve questions of their own.

Building Further: This promotes further transparency, credibility, and encourages further exploration of the topic.

Implications: The potential implications of the findings on vaccine policy, public health, or vaccine development may be further explored, with new data released to the public to increase trust.

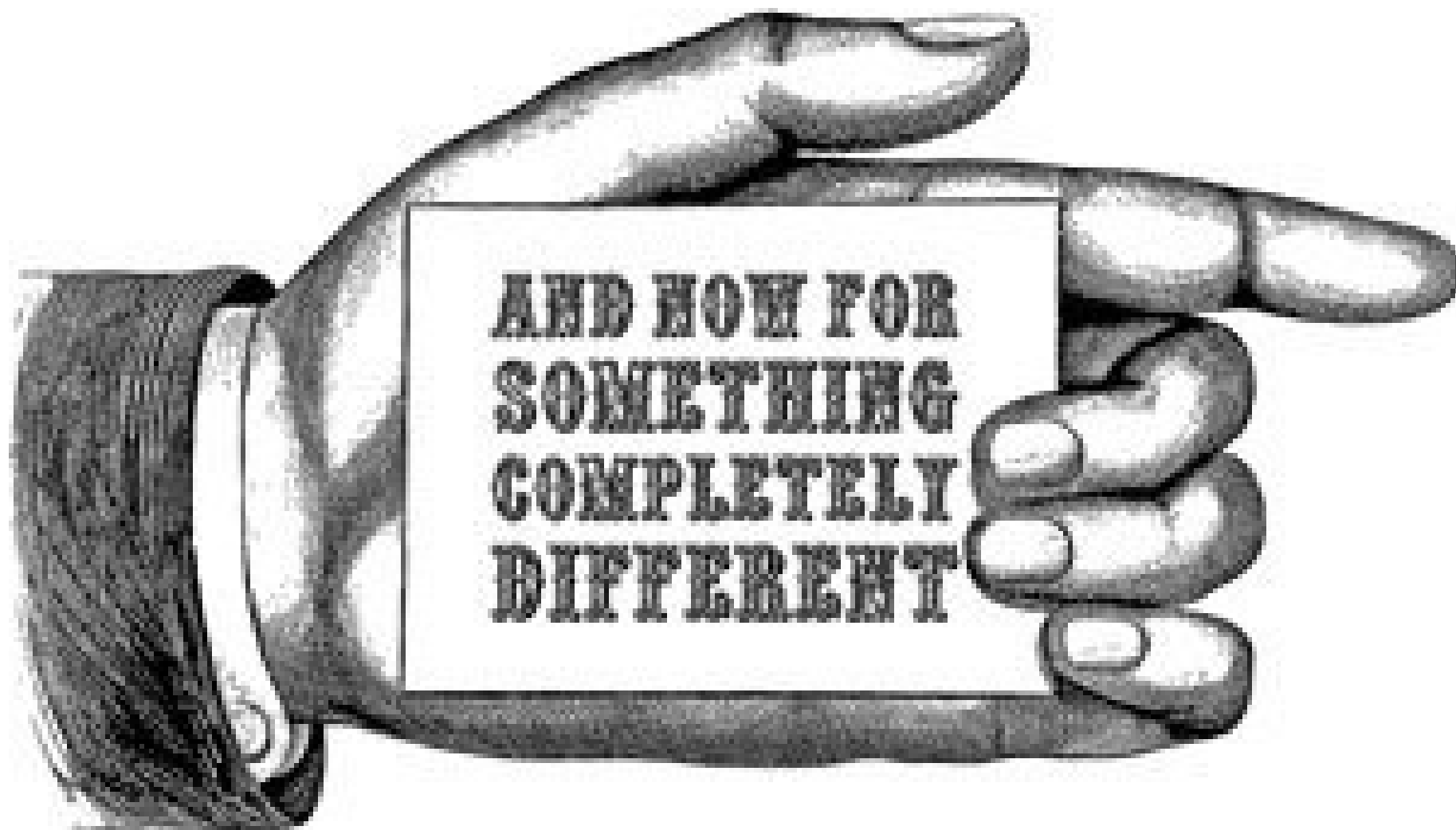
Concluding Ideas

Why is this *also* important?

Furthering Trust: Similar studies may be convinced to also release their data to facilitate investigators to study it and develop deeper understandings of health-related research.

Positive impact: The stakeholders of vaccine education and development would be able to help healthcare providers, patients, and members of communities make informed decisions about vaccine usage.

Let's Talk Databases in R





Relational Databases

- A database table is similar to those that we have been using already.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

attributes
(or columns)

tuples
(or rows)



Let's Look at Some Tables

- `library(tidyverse)`
- `library(nycflights13)`
- `#show built-in tables`

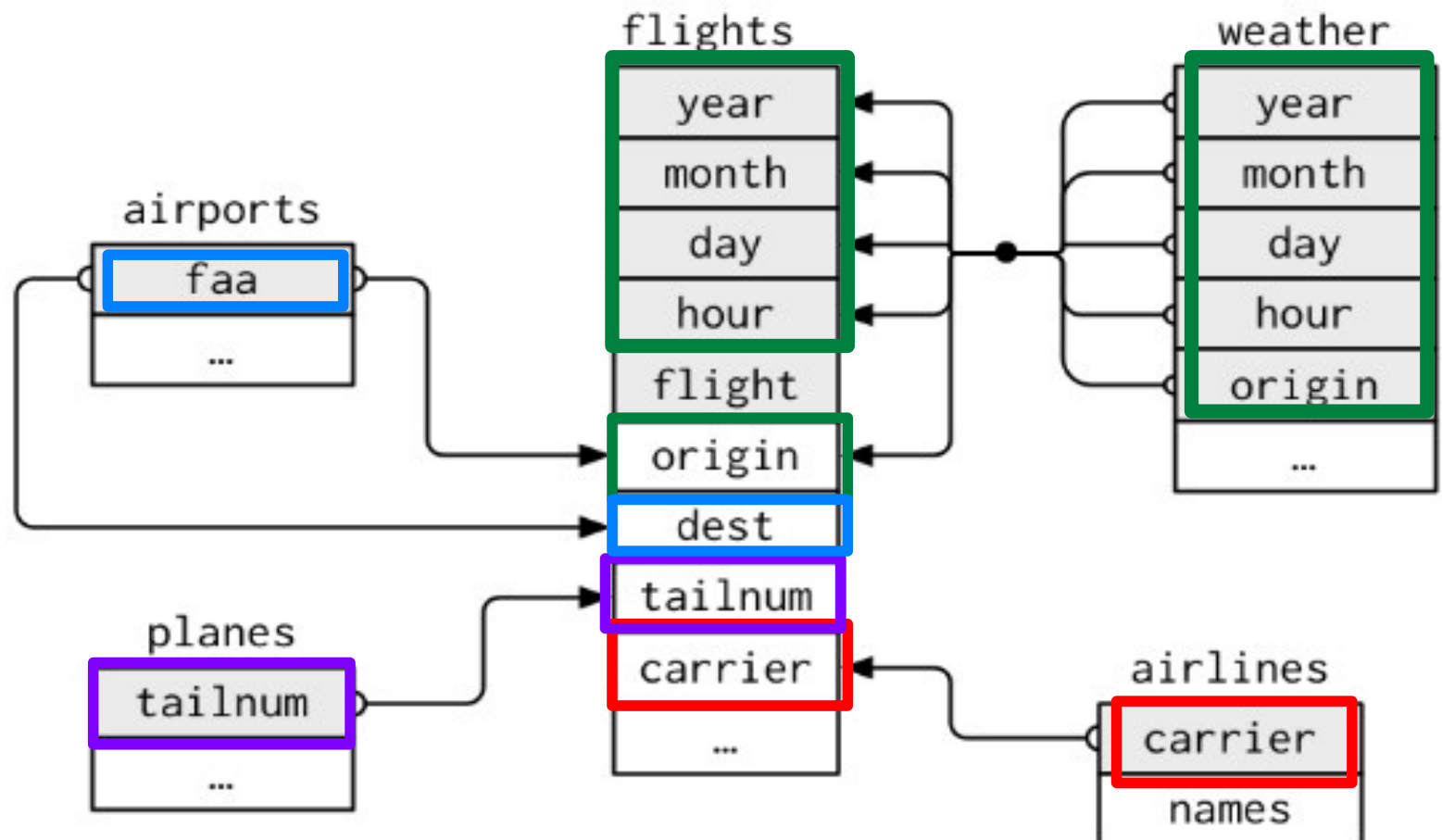
`View(airlines)`

`View(airports)`

`View(planes)`

`View(weather)`

- The data of these built-in tables is “connected” in the style of a database.





Relational Databases

- **Primary Keys:** Unique identifier for each row of the table.
 - Ex: planes\$tailnum
- **Foreign Keys:** Unique identifier for row in another table.
 - Ex: flights\$tailnum
 - Is a foreign key since it exists in the flights table and matches a flight to a unique plane.



Plural or Singular Elements?

If something is unique: there is only one of it. Here each *tailnum* entry is unique. Count the rows

```
planes %>% count(tailnum) %>% filter(n>0)
```

Try setting up a test to see if there are any more than one of an entry (necessary to be a primary key). Count the rows.

```
planes %>% count(tailnum) %>% filter(n > 1)
```

A key could be a combination of things

```
weather %>% count(year, month, day, hour, origin) %>%  
filter(n > 1)
```

```
flights %>% count(year, month, day, flight) %>% filter(n > 1)
```

Note: the $n > 1$ (a tally) comes from the `count()` function

Keys From Singular Elements

- **Baby-name Data**
- First: `install.packages("babynames")`
- `library(babynames)` and `tidyverse`
- Then find the primary keys in, `babynames:babynames`
- **Baseball data:**
- First: `install.packages("Lahman")`
- `library(Lahman)`
- Then find the primary keys in, `Lahman::Batting`



Possible Solutions: Find Some Keys!

- **Baby-name Data**

```
babynames::babynames %>% count(name,  
year, sex) %>% filter(n > 1)
```

- **Baseball data:**

```
Lahman::Batting %>% group_by(playerID,  
yearID, stint) %>% filter(n() > 1) %>% nrow()
```



Simple Analysis with BabyNames

```
library(babynames)
```

```
library(tidyverse)
```

```
# Try this: combine and count common name, year, sex details,  
how many are there for each name?
```

```
babynames::babynames %>% count(name, year, sex) %>%  
filter(n > 1)
```

```
bn <- babynames::babynames %>% select(name)
```

```
# find names beginning with O
```

```
bn[grepl("^O", bn$name),]
```

```
# how many Olivers are there?
```

```
count(bn[grepl("^Oliver", bn$name),])
```